BOUBAKAR DIALLO

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EDUCATION

Cornell University — **B.S. Electrical & Computer Engineering** — **May 2028**

Honors Community Leadership Award - Cornell University (Aug 2025)

Coursework Computer Systems Programming, Computer Architecture, Foundations of Machine Learning, Data Science, Linear Algebra, Embedded Systems, Foundations of Robotics

TECHNICAL SKILLS

Languages Python, C, C++, Java, JavaScript, TypeScript, SQL, Bash

Backend & Cloud FastAPI, REST APIs, Google Cloud Pub/Sub, BigQuery, PostgreSQL, Docker, GCP, dbt

Machine Learning PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Time-Series Forecasting, Computer Vision, NLP

Systems & Tools Linux, Git/GitHub, Data Pipelines, Model Evaluation, Structured Logging, Automated Monitoring

WORK EXPERIENCE

Software Engineer Intern — *Simulacrum, NYC, NY* — **Jan – Jun 2026**

- Engineered distributed real-time data infrastructure with FastAPI and Google Cloud Pub/Sub, processing high-throughput sensor and time-series streams with sub-100 ms end-to-end latency.
- Architected REST APIs, BigQuery storage, and GCP/dbt transformation pipelines with schema evolution and automated data-quality validation, accelerating development approximately 3×.
- Built production observability with structured logging and automated alerts for latency spikes, data drift, and pipeline failures, reducing debugging cycles approximately 30%.
- Evaluated time-series foundation models on 10 years of MLB data, with TiRex and TimesFM achieving 46–54% lower MAE than published regression and neural-network baselines.

Robotics & Hardware Systems Lead — *Engineers Without Borders, Ithaca, NY* — **Oct 2024 – Jan 2026**

- Trained and deployed CNN models on NVIDIA Jetson Orin Nano using transfer learning, spatial augmentation, and cross-validation, improving crop-disease classification accuracy by 40%.
- Developed autonomous data-collection software integrating ROS-controlled robotic arms, drone autopilot firmware, and custom image-capture pipelines across three field sites.

Embedded Systems Intern — *ADAS SAFE, Jacksonville, FL* — **Jun – Aug 2025**

- Developed sensor-validation pipelines transforming radar, IMU, and LiDAR telemetry into structured time-series features for perception-model evaluation.
- Automated anomaly detection using statistical process control and threshold-based classification, reducing sensor alignment errors by 15%.

Incoming Technology Intern — *PNC Bank* — **Summer 2027**

PROJECTS

ML Forecasting Research — *TempusBench* — **Mar – May 2026**

- Built data-processing pipelines aggregating 30M+ NYC ride-hailing records into multivariate time-series datasets spanning 263 city zones, supporting large-scale demand forecasting experiments.
- Diagnosed and patched open-source forecasting codebases using Python and Bash, resolving dependency conflicts and configuration failures to stabilize GPU-based evaluation workflows.
- Benchmarked zero-shot time-series foundation models, including Kronos 2, TiRex, and TimesFM, using NVIDIA T4 GPUs and reproducible evaluation pipelines.
- Achieved 0.26 MAE with TiRex in the reported evaluation and analyzed performance against a published stacking-ensemble baseline reporting 3.337 MAE.

WholeStory Health — *Google Gemini Hackathon Winner* — **May 2026**

- Built an offline multilingual NLP pipeline integrating Whisper speech recognition, language identification, neural translation, and spaCy medical NER, converting patient speech into structured clinical data with 2-second latency.
- Quantized and GPU-optimized models for NVIDIA Jetson Orin Nano deployment, reducing memory footprint by 40% while maintaining inference accuracy.

Signal & Image Classification Pipelines — *Data Science Project* — **Jan 2026**

- Engineered mel-spectrogram, MFCC, and spectral features from 20,000+ audio recordings, applying imbalance handling to achieve 98% classification accuracy.
- Developed satellite-image classification pipelines using HSV and near-infrared features, reducing feature dimensionality by 60% while outperforming the evaluated pixel-based CNN baselines.